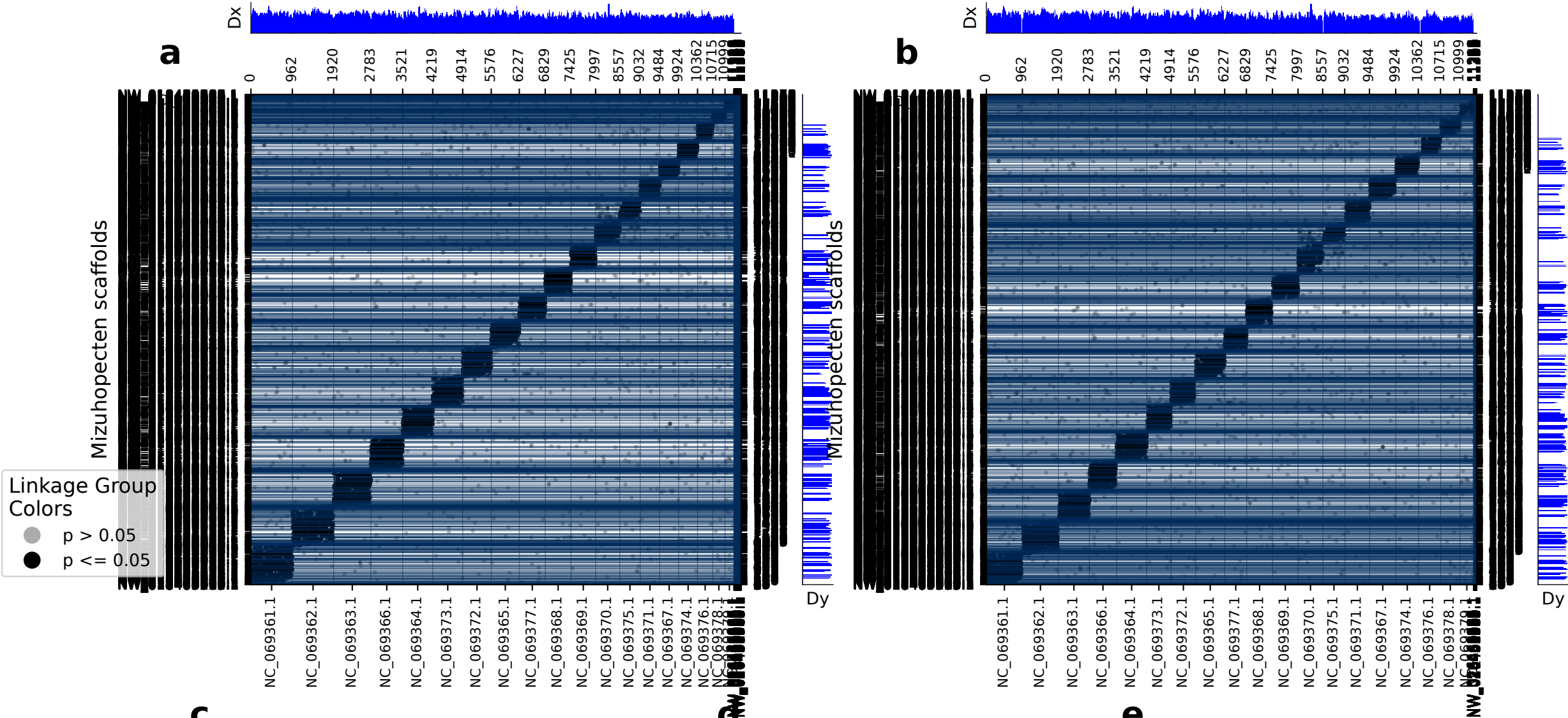


diamond: 11377 orthologs on 176 Mercenaria scaffolds and 1565 Mizuhopecten scaffolds.

Mercenaria vs Mizuhopecten synteny (gene indices)Mercenaria vs Mizuhopecten synteny (chromosome coordinates)



c Mercenaria vs Mizuhopecten scaffolds **d** Mercenaria vs Mizuhopecten FET, whole chromosome **e** Mercenaria vs Mizuhopecten FET, chrom. pieces

	Mercenaria scaf	Mizuhopecten scaf	p (FET)	Count
0	NC_069368.1	NW_018404492.1	1.80e-93	88
1	NC_069366.1	NW_018408902.1	6.97e-66	66
2	NC_069368.1	NW_018407390.1	7.81e-63	60
3	NC_069362.1	NW_018405386.1	1.03e-52	59
4	NC_069369.1	NW_018408896.1	8.65e-64	57
5	NC_069365.1	NW_018404940.1	6.70e-54	50
6	NC_069366.1	NW_018407065.1	1.62e-47	46
7	NC_069366.1	NW_018480077.1	1.44e-40	45
8	NC_069369.1	NW_018408713.1	3.75e-50	44
9	NC_069364.1	NW_018406397.1	5.34e-44	44
10	NC_069364.1	NW_018407700.1	3.49e-38	42
11	NC_069361.1	NW_018405944.1	1.28e-35	42
12	NC_069373.1	NW_018408653.1	2.67e-40	40
13	NC_069369.1	NW_018404097.1	8.01e-42	40
14	NC_069377.1	NW_018408465.1	1.49e-36	39
15	NC_069372.1	NW_018408629.1	1.29e-33	39
16	NC_069363.1	NW_018403543.1	3.01e-34	38
17	NC_069377.1	NW_018404481.1	5.59e-43	37
18	NC_069375.1	NW_018404269.1	3.09e-40	37
19	NC_069374.1	NW_018408252.1	6.30e-37	37
20	NC_069372.1	NW_018408761.1	1.69e-36	37
21	NC_069369.1	NW_018407435.1	3.65e-37	37
22	NC_069363.1	NW_018408899.1	3.10e-34	37
23	NC_069361.1	NW_018408895.1	2.21e-31	36
24	NC_069366.1	NW_018404662.1	2.99e-25	35
25	NC_069363.1	NW_018480076.1	3.04e-33	35
26	NC_069363.1	NW_018404373.1	1.24e-33	34
27	NC_069362.1	NW_018407509.1	1.77e-26	34
28	NC_069361.1	NW_018408638.1	5.54e-32	34
29	NC_069373.1	NW_018480078.1	2.65e-29	33
30	NC_069372.1	NW_018408154.1	2.82e-29	33
31	NC_069369.1	NW_018406415.1	1.88e-29	33
32	NC_069364.1	NW_018406060.1	7.98e-32	33
33	NC_069363.1	NW_018408645.1	7.12e-30	32
34	NC_069361.1	NW_018408887.1	2.50e-28	32
35	NC_069374.1	NW_018408921.1	2.56e-33	31
36	NC_069375.1	NW_018408916.1	1.71e-24	30
37	NC_069374.1	NW_018408742.1	1.13e-36	30
38	NC_069377.1	NW_018405900.1	1.38e-32	29
39	NC_069377.1	NW_018405342.1	5.91e-29	29
40	NC_069376.1	NW_018408880.1	2.83e-29	29
41	NC_069374.1	NW_018407426.1	4.56e-34	29
42	NC_069365.1	NW_018407364.1	5.95e-28	29
43	NC_069374.1	NW_018403589.1	2.63e-27	28
44	NC_069373.1	NW_018408623.1	6.45e-27	28
45	NC_069368.1	NW_018406874.1	2.05e-31	28
46	NC_069367.1	NW_018408726.1	7.55e-28	28
47	NC_069366.1	NW_018406970.1	2.59e-24	28
48	NC_069366.1	NW_018406089.1	4.02e-22	28
49	NC_069363.1	NW_018407733.1	2.14e-22	28
50	NC_069373.1	NW_018403786.1	7.48e-27	27
51	NC_069372.1	NW_018404519.1	2.70e-26	27
52	NC_069368.1	NW_018408493.1	1.06e-25	27
53	NC_069365.1	NW_018408919.1	1.61e-25	27
54	NC_069364.1	NW_018405113.1	4.69e-21	27
55	NC_069363.1	NW_018406690.1	7.71e-20	27
56	NC_069362.1	NW_018404995.1	6.00e-21	27
57	NC_069371.1	NW_018408820.1	1.22e-27	26
58	NC_069370.1	NW_018408786.1	2.10e-24	26
59	NC_069366.1	NW_018405747.1	7.13e-23	26
60	NC_069366.1	NW_018404308.1	7.13e-23	26
61	NC_069361.1	NW_018407511.1	2.64e-23	26
62	NC_069366.1	NW_018408845.1	9.15e-24	25
63	NC_069364.1	NW_018408914.1	2.50e-22	25

(a) depicts the chromosomes/scaffolds of Mercenaria (x-axis) plotted against the chromosomes/scaffolds of Mizuhopecten (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11377 orthologs found between 176 Mercenaria scaffolds and 1565 Mizuhopecten scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visually identifying gene-poor regions of the chromosomes (ends) and gene-rich regions (middle). (c) shows the FET p-values for the correlation between the chromosome pair and the gene group. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a p-value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the FET p-values for the correlation between the chromosome pair and the gene group. Useful for showing if single arms of chromosomes are correlated with other regions.